

Diesel engine

16V4000G60

Spare parts catalog for product no.

527101155

Printed in Germany

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Preface

This catalog is published for parts requisitioning purposes and provides assistance in engine disassembly and reassembly. It contains a complete list of all engine components, including those which are not available as replacement parts.

The parts originate from the Engine Design File and document the product's construction status.

Catalog Arrangement

The catalog is subdivided into functional groups and functional group segments (FG/FGS).

The headings of the illustration pages contain the following information:

- ◆ **TYPE**
- ◆ **SERIAL NO.**
- ◆ **FG/FGS DESTINATION**
- ◆ **FG/FGS NO.**
- ◆ **BOM NO.**

Explanation of Columns

- ◆ **POS** Provides a cross-reference from illustration to text section. Non-illustrated parts are indicated by the suffix "Z" after the C/N.
- ◆ **PART NO.** This is the valid number for ordering the associated part.
- ◆ **DESIGNATION** The precise part name.
- ◆ **QTY (Quantity of Unit)** This column indicates the quantity of this part required for the functional group.
- ◆ **MEASUREMENT**
- ◆ **FN (Footnote)** Provides additional parts information. For clarification of footnote symbols, see Page 3/2.
- ◆ **REMARKS** This column is intended for supplementary information.

Amendments Service

Part No. changes and amendments which occur after publication of the catalog will not be published. We therefore recommend updating of the catalog part numbers by the end user. This information is provided, for instance, in order confirmations and/or on equipment labels.

Footnote Symbols Identification

AB Supplied as a complete assembly. Repair by manufacturer only.

AC Not scheduled as a spare part. Availability must be checked.

EM Emission-relevant part. In the case of all marine engines with IMO certification it should be noted that the identification numbers of the emission-relevant parts on the engine must match the details given in the technical file. When ordering, quote EM code with order no. E.g: 535 030 04 17 EM.

KD Group balancing required.

KE Individual parts not obtainable through the parts supply system.

KF Calibrated part.

M Parts having different repair sizes. Dimensions are listed in the Tolerances and Wear Limits List.

P Bought-out components.

SB Group of parts which are welded or brazed together.

UA Indicate dimensions when ordering.

UB Only supplied complete or as a complete set.

Ordering Procedure

The following information is essential for spare-part order processing:

- ◆ **TYPE**
- ◆ **SERIAL NO.**
- ◆ **FG/FGS**

- ◆ POS
- ◆ PART NO.
- ◆ DESIGNATION
- ◆ QUANTITY
- ◆ PAGE

When ordering replacement parts for ancillary equipment such as injection pumps, fuel pumps, etc., give both model and part number of the part.

Phone orders should be confirmed in writing to prevent errors.

Ordering Example

Type	Serial number	FG/FGS	BOM	POS.	Part number	Designation	Qty.	P
8V4000G60	524100298	075 05	070.00075	1	0010104351	Injector	8	
		083 05	524.903-30	10	5240900252	Fuel Filter	2	

Factory Repairs

Parts shipped to MTU for repair should have an engine serial-number identification (tag or label, for example).

Exchange Components (Reman)

Exchange components are parts overhauled by the manufacturer, providing the same properties with regard to service life and quality as new parts.

Components which are included in the Reman exchange procedure, are marked by the following letters added before the item number:

E 524 010 35 20 = New component in exchange procedure.

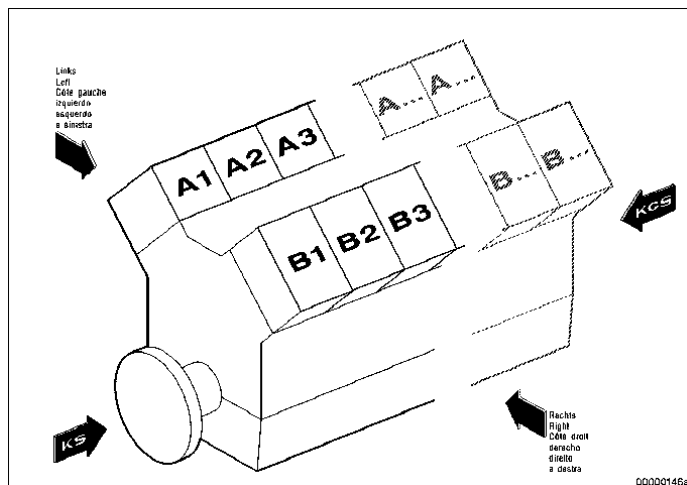
R 524 010 35 20 = Reman component in exchange procedure.

Exchange Procedure

With the introduction of exchange components, the so-called core deposit will also be charged for new parts. The core deposit will be indicated separately on the invoice and the respective value will be credited upon return of reusable components.

For further information refer to Service Information 01/S0028.

Engine Side And Cylinder Designations



Engine sides are always designated as viewed from driving end.

The left bank of cylinders is marked 'A' and the right bank 'B' (according to DIN ISO 1204).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the drive end.

Numbering of other engine components is also from the drive end, starting with No. 1.

Designations and Abbreviations

Driving end = KS

Free end = KGS

Left side
Right side
Upper side
Lower side
Turbocharger = ATL

Overview functional groups

FG FG DESIGNATION

004 LABELS AND MISCELLANEOUS MATERIALS

011 CRANKCASE

013 CYLINDER LINER

014 OIL PAN

015 ENGINE MOUNTING

016 END HOUSING, MAIN PTO END AND AUX PTO END

018 CRANKCASE BREATHER

024 GEARCASE, AUX PTO END

025 GEAR TRAIN, AUX PTO END

031 CRANKSHAFT ...

032 CRANKSHAFT SUPPORT BEARINGS, MAIN PTO END

035 CRANKSHAFT, PTO, AUX PTO END

036 CRANKSHAFT TRANSPORTATION LOCKING DEVICE

037 PISTON

038 CONROD

041 CYLINDER HEAD

042 CYLINDER HEAD ATTACHMENTS

051 CAMSHAFT

052 CAMSHAFT DRIVE

053 CAMSHAFT AXIAL ALIGNMENT BEARING

054 ROLLER TAPPETS, PUSH RODS

055 VALVE DRIVE

073 INJECTION PUMP / HP PUMP

075 FUEL INJECTOR

077 HP FUEL LINE / HP LINE

081 FUEL PIPEWORK BEFORE DELIVERY PUMP

083 FUEL FILTER

084 FUEL SUPPLY TO INJECTION PUMP

086 LEAK-OFF FUEL LINE

101 TURBOCHARGER (H.P)

111 CHARGE AIR COOLING, IF REQUIRED, LEFT SIDE

121 AIF FILTER

124 AIR SUPPLY TO CYLINDERS

125 AIR SYSTEM, TURBOCHARGER - INTERCOOLER

141 EXHAUST SYSTEM, CYLINDER HEAD - EXH. MANIFOLD

172 ELECTRIC STARTING

181 LUBE OIL PUMP AND DRIVE

183 OIL FILTRATION/COOLING

184 CRANKCASE-INTERNAL OIL SUPPLY

185 TURBOCHARGER OIL SUPPLY

202 HIGH-TEMPERATURE CIRCUIT

203 LOW-TEMPERATURE CIRCUIT

206 COOLANT PIPEWORK WITH THERMOSTAT

231 ENGINE MOUNTING

507 GENERAL WIRING, ENGINE/GEARBOX/PLANT

508 ENGINE GOVERNOR (ELECTRONIC) CONTROL SYSTEM

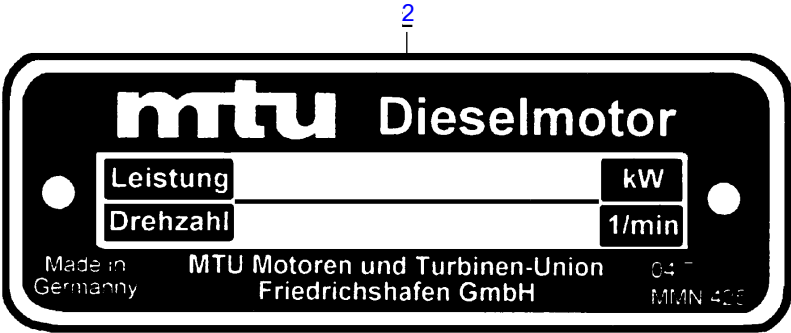
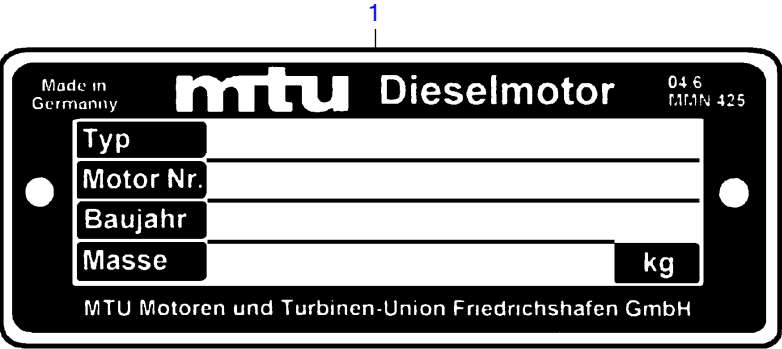
511 ALARM SYSTEM

GROUP NUMBER (FG) GROUP SEGMENT (FGS)
GROUP NAME
BOM NO.

004 05
INSTALLATION OF LABELS
000.00115

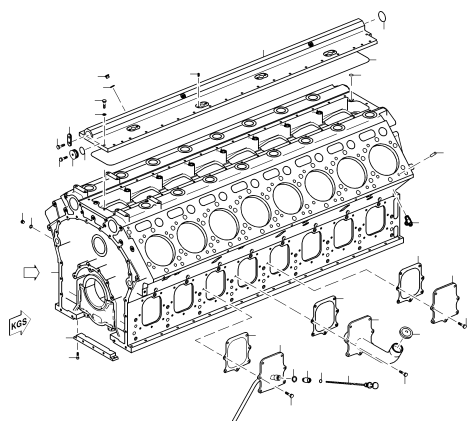


00400006a



00400006a

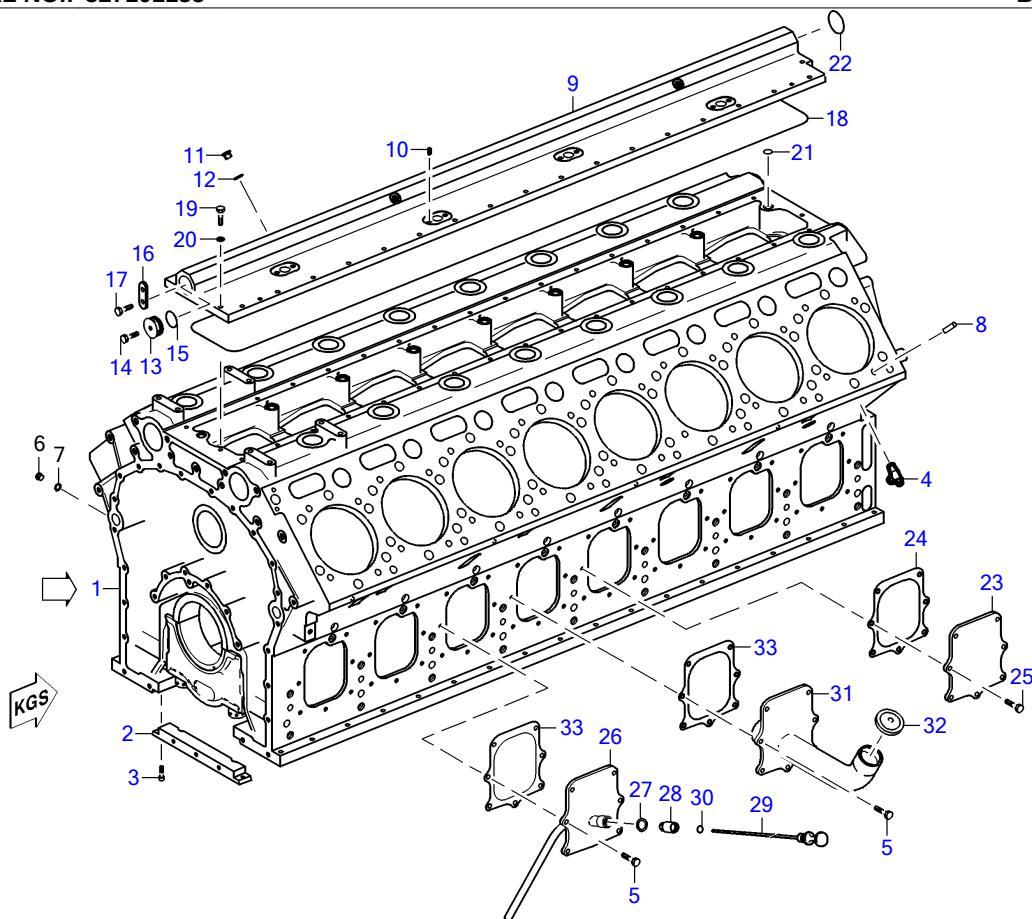
POS.	PART NO.	DESIGNATION	QTY.	MEASUREMENT
1	700425000135	PLATE	1	04.6
2	700425000136	PLATE	1	04.7



01100012a

011
CRANKCASE
527.903-30

05



01100012a

POS.	PART NO.	DESIGNATION	QTY.	MEASUREMENT	FN.
1	5270100605	CRANKCASE	1		
2	5240110440	STRIP	1		
3	000912008115	SCREW	2	M 8 X 16	
4	0049976736	DRAIN VALVE	2		
8	000007010124	DOWEL PIN	16	10 M6 X 20	
9	5270101214	COVER	1		
10	000835010056	STUD	8	M 10 X 25	
11	000908014001	PLUG	1	M 14 X 1,5	
12	007603014102	SEALING RING	1	A 14 X 18	
13	735084050000	PLUG	1	50	
14	000933010089	SCREW	1	M 10 X 20	
15	700429042000	O-RING	1	42 X 5	
16	5249950059	STRAP	1		
17	000933010001	SCREW	1	M 10 X 10	
18	5279970345	O-RING	1	135X5	
19	000933010093	SCREW	36	M 10 X 35	
20	000125010524	WASHER	38	A 10,5	
21	700429018005	O-RING	10	18 X 4	
22	700429055001	O-RING	1	55 X 5	
23	5240110732	INSPECT PORT COVER	6		
24	5240111380	GASKET	10		